MIL-I-7088A
18 JUNE 1979
SUPERSEDING
MIL-I-7088
29 January 1951

MILITARY SPECIFICATION

INDICATOR, PRESSURE, FUEL, 0-1,000 PSI

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers one type of hermetically sealed, remote indicating, synchro-style fuel pressure indicator.

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

PPP-B-636 - Box, Shipping, Fiberboard

MILITARY

MIL-P-116 - Preservation-Packaging, Methods of
MIL-T-5396 - Transmitter, Pressure, Fuel, 0-1,000 PSI
MIL-I-7057 - Indicator, Synchro, Aircraft, General Specification for

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Hq AFLC CASO/LODS, Federal Center, Battle Creek, MI 49016, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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STANDARDS

MILITARY

MIL-STD-129	- Marking for Shipment and Storage
MIL-STD-130	- Identification Marking of U.S. Military Property
MIL-STD-794	- Parts and Equipment, Procedures for Packaging and Packing of
MS28005	- Transmitter, Pressure, Synchro, Aircraft
MS28010	- Indicator, Pressure Synchro, Single, 2-Inch Size
MS33585	- Pointer, Dial, Standard Design of Aircraft Instrument

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 General. The requirements specified in specification MIL-I-7057 are applicable as requirements of this specification. Additional requirements shall be as specified herein.

3.1.1 General specification. Where the requirements of the general specification and this specification conflict, the requirements of this specification shall govern.

3.2 Design. The indicator shall be designed for use with an MS28005-4 pressure transmitter in accordance with specification MIL-T-5396.

3.3 Construction.

3.3.1 Case. The case shall be hermetically sealed and the case dimensions shall conform to standard MS28010.

3.3.1.1 A separate nonhermetically sealed cap may be used on the back of the case for the purpose of installing the electrical receptacles. The cap shall be made of nonferrous, low-density metal.

3.3.1.2 The distance between the outer surface of the cover glass and the front edge of the case shall be the minimum practicable and shall not exceed 1/32 inch.

3.3.2 Pointer. The pointer shall be in accordance with MS33585-8, except that the length of the pointer shall be such

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that the tip will extend into the scale a distance equal to 1/3 to 2/3 the length of the shortest graduation.

3.3.2.1 Pointer finish. The shaded portion of the pointer shall be finished with fluorescent-luminescent material.

3.3.3 Dial. The dial shall be as shown on figure 1. A minimum diameter of 1-3/4 inches shall be maintained across the outside ends of the graduations. The graduation for 300 psi shall be located on the horizontal center line at the left side of the dial.

3.3.3.1 Dial markings.

3.3.3.1.1 Fluorescent-luminescent markings. The following markings shall be finished in fluorescent-luminescent material. The dimensions of the markings shall be as follows:

	Height or Length Inch ±1/64	Width of Line or Graduation Inch ±0.005
Numerals 0, 1, 2, 4, 6, 8, and 10	5/32	0.025
Graduations at 0, 100, 200, 300, 400, 600, 700, 800, 900, and 1,000	3/16	0.031
Graduation at 500	9/64	0.031
20 psi graduations from 0 to 100	3/32	0.020
50 psi graduations	3/32	0.020
Lettering "FUEL PRESS"	1/8	0.025
Lettering "PSI X 100"	3/32	0.020

3.3.3.1.2 Durable dull black markings. The markings for the applicable MS part number shall be permanently and legibly marked on the dial in letters 1/16 inch high; location is optional. These markings and all other markings not otherwise specified shall be furnished in durable dull black.

3.3.4 Weight. The weight of the indicator shall not exceed 0.60 pound.

3.4 Identification of product. Equipment, assemblies, and parts shall be marked for identification in accordance with MIL-STD-130.

4. QUALITY ASSURANCE PROVISIONS

4.1 The fuel pressure indicator shall be subjected to inspection and tests in accordance with specification MIL-I-7057.

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4.2 Test tolerances. The indicator shall be within the tolerances specified in tables I and II of this specification when tested in accordance with specification MIL-I-7057.

TABLE I

Pressure Indicator

PSI	Tolerance ± psi
0	3.0
60	3.0
100	3.0
200	3.0
300	3.0
400	3.0
500	3.0
600	3.0
700	3.0
800	3.0
900	3.0
1,000	3.0

TABLE II

Pressure Indicator

Test	Tolerance ± psi
Position error at 500 psi	5.0
Friction	
Room temperature	5.0
High temperature	5.0
Low temperature	8.0
Vibration	
Pointer oscillation	5.0
Pointer variation	5.0
Scale error	
Low temperature	5.0
High temperature	5.0

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4.3 Inspection and test. Tests of methods of preservation and packaging shall be accomplished in accordance with section 4 of MIL-P-116, to insure compliance with section 5 of this specification.

5. PACKAGING

5.1 Preservation-packaging. Preservation-packaging shall be level A or C as specified (see 6.2).

5.1.1 Level A.

5.1.1.1 Cleaning. Indicator shall be cleaned in accordance with process C-1 of MIL-P-116.

5.1.1.2 Drying. Indicator shall be dried in accordance with process D-4 of MIL-P-116.

5.1.1.3 Preservation application. Not applicable.

5.1.1.4 Unit packaging. Unless otherwise specified by the procuring activity, each indicator shall be packaged in quantity unit packs of one each in accordance with method IA-8 of MIL-P-116. Overbox in PPP-B-636 carton large enough to allow for application of sufficient cushioning material between container and bag, of a type, density, and thickness to insure shock transmission does not exceed peak values in Gs established for the item, when completed packs are subjected to the rough handling drop tests of MIL-P-116.

5.1.2 Level C. Indicators shall be clean, dry, and individually packaged in a manner that will afford adequate protection against corrosion, deterioration, and physical damage during shipment from supply source to the first receiving activity.

5.2 Packing. Packing shall be level A, B, or C as specified (see 6.2).

5.2.1 Level A. Indicators packaged as specified in 5.1.1 shall be packed in shipping containers conforming to PPP-B-636, weather-resistant class, unless otherwise specified by the procuring activity. Insofar as practical, exterior shipping container shall be of uniform shape, size, of minimum tare and cube consistent with the protection required. Table II of MIL-STD-794 applies for Army use.

5.2.2 Level B. Indicators packaged as specified in 5.1.1 shall be packed in domestic class exterior shipping containers conforming to PPP-B-636. Other requirements as specified in 5.2.1 above shall apply.

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5.2.3 Level C. Packing shall be applied which affords adequate protection during domestic shipment from the supply source to the first domestic receiving activity for immediate use.

This level shall conform to applicable carrier rules and regulations and may be the contractor's commercial practice provided the latter meets the requirements of this level.

5.3 Marking of shipments. Interior packages and exterior shipping containers shall be marked in accordance with standard MIL-STD-129. The nomenclature shall be as follows: Indicator, Pressure, Fuel, 0-1,000 PSI, specification MIL-I-7088, Manufacturer's Part No. *____.

*Applicable data to be entered by the contractor.

5.3.1 In addition, the following marking shall be added for shipments to the Air Force only:

IF IN STORAGE AFTER (DATE)[1], THE INDICATOR SHALL BE TESTED AND INSPECTED BEFORE ISSUE

1 This date shall be 18 months from the date of inspection.

6. NOTES

6.1 Intended use. The fuel pressure indicator covered by this specification is intended for use in conjunction with a suitable transmitter in indicating remotely the pressure in the fuel systems of aircraft engines.

6.2 General. The instruction contained in section 6 of specification MIL-I-7057 are applicable to and form a part of this specification.

6.3 Reclaimed materials. The use of reclaimed materials shall be encouraged to the maximum extent possible.

6.4 Qualification. With respect to products requiring qualification, awards will be made only for such products as have, prior to the time set for opening of bids, been tested and approved for inclusion in the applicable Qualified Products List, whether or not such products have actually been so listed by that date. The attention of the contractors is called to this requirement, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be covered by this specification. The activity responsible for the Qualified Products List is Hq AFLC CASO/LODS, Federal Center,

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Battle Creek, MI 49016, and information pertaining to qualification of products may be obtained from that activity.

Custodian:
Air Force - 99
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Preparing Activity:
Air Force - 99

Project Number:
6620-0323

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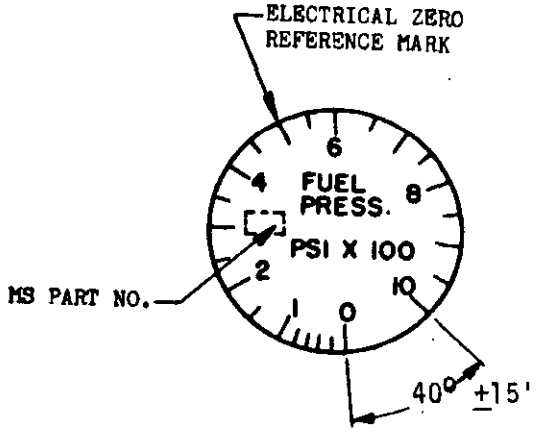


FIGURE 1. Dial.

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
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DOCUMENT IDENTIFIER AND TITLE

MIL-I-7088A

NAME OF ORGANIZATION AND ADDRESS

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MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?

☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

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